From Tunneling Junctions to Continuum Response: A Physics-Informed Computational Model for Strain-Dependent Conductivity in Piezoresistive Systems

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Abstract

Many sensing devices rely on transduction mechanisms that convert mechanical deformation into measurable electrical signals. One important example is the piezoresistive effect, in which the electrical conductivity of a material varies with mechanical strain. This coupling between mechanical deformation and electrical conduction provides a natural setting for the study of multiphysics systems.

This thesis develops a mathematical model describing conductive materials whose electrical conductivity depends on the underlying mechanical strain field. The mechanical behavior is described by the equations of elasticity, while the electrical potential satisfies a conductivity equation with strain-dependent coefficients. The resulting formulation leads to a nonlinear coupled system of partial differential equations of the form

$$\nabla \cdot \boldsymbol{\sigma}(u) = 0, \qquad \nabla \cdot (\sigma(\varepsilon(u))\nabla V) = 0,$$

where u denotes the displacement field, V the electrical potential, and the conductivity coefficient depends on the strain tensor $\varepsilon(u)$.

The primary objective of this work is to investigate this coupled electromechanical system from a computational modeling perspective. Emphasis is placed on the formulation and analysis of the underlying multiphysics model, as well as on the development of computational strategies for approximating its solutions. In addition to classical numerical techniques, the study explores hybrid approaches that combine physics-based modeling with data-driven approximation methods to represent complex strain–conductivity relationships.

The proposed framework aims to provide insight into strain-dependent electrical behavior in deformable media and contributes to the broader study of nonlinear multiphysics PDE systems arising in sensing and material modeling.

The results of this study are expected to provide insight into strain-dependent electrical behavior in deformable materials and contribute to the design and analysis of piezoresistive systems. Potential applications include structural health monitoring, wearable electronics, and soft robotics.

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1 Introduction

The piezoresistive effect arises from the coupling between mechanical deformation and electrical conduction. In typical applications, a doped semiconductor element (such as silicon) is embedded within a mechanical structure, where external forces induce strain in the material. This strain alters the electrical conductivity, which can be measured through changes in voltage.

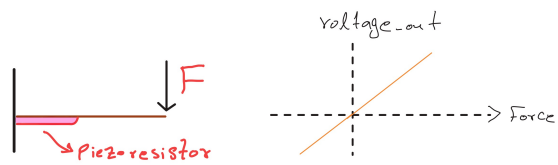

Figure 1: A schematic representation of the piezoresistive effect, where mechanical deformation leads to changes in electrical response.

The piezoresistive effect, first observed by William Thomson (Lord Kelvin) in the 19th century and later developed for semiconductors by Bardeen, Shockley, and Smith [3], describes the dependence of electrical conductivity on mechanical deformation.

There exists a wide spectrum of engineering manifestations of the piezoresistive effect, including strain gauges, pressure sensors, microelectromechanical systems (MEMS), cantilever-based sensing devices, and flexible or wearable systems [1].

While these systems differ in geometry, material composition, and boundary conditions, they can be interpreted as different realizations of a common underlying multiphysics system.

Despite extensive technological development, the electromechanical coupling is often treated in an application-specific manner. This motivates the need for a unified mathematical framework that captures the essential interaction between mechanics and electrical conduction.

2 Problem Statement and Motivation

Consider a conductive element with length L, cross-sectional area A, and resistivity ρ. Its electrical resistance is given by

$$R = \frac{\rho L}{A}. \tag{1}$$

Under mechanical loading, both geometry and material properties change. In particular, the length, area, and resistivity become functions of deformation, making resistance dependent on the mechanical state.

Using Ohm's law,

$$R = \frac{V}{I}, \tag{2}$$

and expressing voltage and current as

$$V = EL, \qquad I = JA, \tag{3}$$

we obtain

$$\frac{EL}{JA} = \frac{\rho L}{A}. \tag{4}$$

Canceling geometric terms yields the local relation

$$\mathbf{J} = \sigma \mathbf{E}, \qquad \sigma = \frac{1}{\rho}, \tag{5}$$

where $\mathbf{E} = -\nabla V$.

Combining this with charge conservation in steady state,

$$\nabla \cdot \mathbf{J} = 0, \tag{6}$$

leads to the conductivity equation

$$\nabla \cdot (\sigma \nabla V) = 0. \tag{7}$$

The resulting equation is an elliptic PDE in divergence form, belonging to a well-studied class of second-order equations in the theory of partial differential equations [2].

In piezoresistive materials, the conductivity depends on strain:

$$\sigma = \sigma(\varepsilon(u)), \tag{8}$$

where the strain tensor is given by[1]

$$\varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T). \tag{9}$$

This introduces a nonlinear coupling between mechanical deformation and electrical conduction.

This illustrates that both geometric deformation and intrinsic material changes contribute to the variation of resistance.

3 Mathematical Model

The coupled electromechanical system is described by

$$\nabla \cdot \boldsymbol{\sigma}(u) = 0, \tag{10}$$

$$\nabla \cdot (\sigma(\varepsilon(u)) \nabla V) = 0, \tag{11}$$

where u is the displacement field and V is the electrical potential.

This system captures the interaction between elasticity and electrical conduction in a continuum setting.

[1] The strain tensor is defined as the symmetric part of the displacement gradient, which removes rigid body rotations and captures the true deformation of the material.

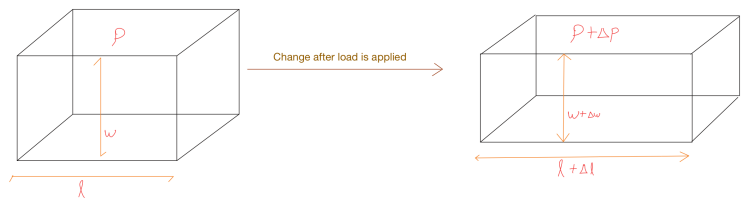

Figure 2: Change of geometric and material properties under mechanical deformation. L denotes the length of the conductor, W its characteristic width, and $A = W^2$ the cross-sectional area (assuming a square cross-section). The material property ρ represents the electrical resistivity, which may vary under mechanical deformation. Both the dimensions and the resistivity of the conductor vary, leading to a strain-dependent resistance.

4 State of Research

The piezoresistive effect has been extensively studied [3, 1] in experimental and engineering contexts, particularly in sensor design and microelectromechanical systems (MEMS). Classical works established the fundamental behavior of strain-dependent conductivity in semiconductors, while more recent research focuses on device optimization and fabrication techniques.

However, most existing approaches treat the electromechanical coupling in an application-specific manner. From a mathematical perspective, coupled PDE systems with state-dependent coefficients are well understood in general, but their systematic application to piezoresistive materials remains limited. This motivates the development of a unified computational framework capable of capturing the underlying multiphysics behavior.

Based on these considerations, the proposed study is guided by the following research questions:

- How can strain-dependent conductivity be incorporated into a consistent continuum PDE model?
- How does mechanical deformation influence the distribution of electrical potential?
- What are the effects of nonlinear coupling on system behavior and sensitivity?
- To what extent can data-driven approaches improve the modeling of $\sigma(\varepsilon)$?

To address these questions, the research will be structured around the following key components:

- Introduction and physical background
- Derivation of the coupled electromechanical model
- Analysis of the nonlinear PDE system
- Computational methods and numerical experiments
- Results and discussion

5 Methodology

The study begins with the formulation of the coupled PDE system within a continuum mechanics framework.

Numerical methods are employed to approximate the solution, with particular attention to handling the nonlinear coupling between mechanics and electrical conduction.

In addition, hybrid approaches are explored, where the constitutive relation $\sigma(\varepsilon)$ is approximated using data-driven models.

The methodology aims to balance physical accuracy with computational efficiency.

6 Expected Outcome

The expected outcome is a computational framework for analyzing strain-dependent electrical conduction.

The model will provide insight into how deformation affects electrical response and sensitivity.

From a broader perspective, the work contributes to the study of nonlinear multiphysics PDE systems and may support applications in sensing, materials modeling, and structural monitoring.

References

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